

Anthony Ibarra

Embedded Systems Engineer

(Local to Chicago, IL; Available to Start Onsite)

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Professional Summary:

Electrical and Computer Engineering graduate looking for the opportunity to become a professional in my respective field. Showcasing a strong focus in Embedded Systems, Control system theory, Robotics and Automation. With fundamental understanding of pattern recognition, Signal processing, and power electronics making me a flexible Embedded & Control Systems Engineer. Possessing various knowledge in different fields providing fast adaptable capability within the work environment of a multidisciplinary team. With the eagerness to learn on and off the job, demonstrating reliability and trustworthiness for a great addition toward a multidisciplinary team.

Education:

University of Illinois Chicago (UIC)

Jan 2023 - May 2025

Master of Science in Electrical and Computer Engineering Chicago, United States

Relevant Coursework:

- Convex Optimization, Mechatronics Embedded Design, Neural Networks
- Audio & Acoustic Signal Processing, Filter Synthesis, Adaptive Digital Filters, Electromechanical Energy Conversion
- Linear Systems Theory & Design, Electromagnetic Compatibility, Advanced Computer Communication

University of Illinois Chicago (UIC)

Aug 2017 - May 2022

Bachelor of Science in Computer Engineering Chicago, United States

Relevant Coursework:

- Artificial Intelligence, Principles of Modern Control, Principles of Auto Control, Pattern Recognition
- Computer Comm Networks, Robotics: Algorithm/Control, Embedded Systems

Certifications:

- Creating and Securing Bluetooth Low Energy (BLE) Application - LinkedIn Learning
- Advanced 5G NR (New Radio) Technologies - LinkedIn Learning
- Satellite Internet and Communications: Fundamentals - LinkedIn Learning
- C Programming for Embedded Applications - LinkedIn Learning
- Learning FPGA Development - LinkedIn Learning
- Learning Verilog for FPGA Development - LinkedIn Learning
- Introduction to FreeRTOS and Basic Task Management - LinkedIn Learning
- Getting Started with RISC-V - LinkedIn Learning
- Learning LabVIEW - LinkedIn Learning
- Data Acquisition with LabVIEW - LinkedIn Learning
- Robot Framework Test Automation: Level 1 (Selenium) - LinkedIn Learning

Skills:

- C, C++, Python, Bash, MIPS, ARM, FPGA, Verilog, VHDL
- BLE, UDP, TCP/IP, I2C, UART, SPI, Ethernet, RF, LTE
- Raspberry PI, Arduino IDE, assembly language
- HIL, MATLAB, Simulink, Quartus, Icarus Verilog
- Wireshark, Robotics, ROS2, Ignition, Rockwell, PLC, SCADA, HMI
- Altium, ANSYS HFSS, LT Spice, PCB, soldering, SolidWorks
- Git, GitHub, Docker, IEEE, OOP, IoT, Ubuntu, MySQL
- Linux, Ubuntu, Visual Studio Code, VNC
- Latex, Microsoft PowerPoint Microsoft Word

Professional Experience:

University of Illinois Chicago (UIC) Chicago, United States

Jan 2025 – May 2025

Research Project, Acoustic & Audio Signal Processing

- Implement a solution for motorcycle riders with some degree of hearing impairment.
- The starting goal is to remove the dampening from the helmet, providing clear sound while wearing the helmet without adding any unneeded noise or unwanted sounds.
- In turn this can potentially amplify user awareness while riding or in largely populated areas. Report being in the

scope and style of a Signal Processing Society conference paper.

- Develop a new method or solution with concepts to be applied to an existing problem.
- Perform real world testing and validation of project and research implementation.
- Test and use skills and algorithms from acoustic & audio signal processing to achieve the set goal

University of Illinois Chicago (UIC) Chicago, United States

Jan 2024 - May 2024

Optimization on Three Coil Long Range (WPT)

- Electromagnetic Compatibility testing using a spectrum analyzer, Oscilloscopes for EMC and EMI testing for PCB and signal integrity using RF SMA coaxial cables, SMA terminators
- Create a three-coil configuration for long range wireless power transfer for simulation, confirming data from research papers through simulations
- Mathematically simulated plot from MATLAB regarding the Three coils as shown within the research paper
- Simulation and plot from Ansys HFSS 3D EM simulation software, simulating the three-coil behavior in different configurations
- Make an IEEE formatted report using Latex, in a professionally typeset manner demonstrating discoveries and results

University of Illinois Chicago (UIC) Chicago, United States

Jan 2023 - May 2023

Self-Driving Car, Mechatronics Embedded Design

- Lead the team and manage all time constraint tasks, development and design tasks, make timely decisions for the team's success in completing the product and hardware development
- Design a motor driver circuit through FET Driver implementation, either of (single fet, half bridge, or full H-bridge), controlled via a PWM input signal from the microcontroller
- Design a Boost Converter for the implementation multiple power systems
- Product testing using Oscilloscope, multimeters, benchtop powersupplies; and Soldering fixes when necessary
- Develop and tune PID controllers for both the steering and velocity controls (Servo motor, DC motor) respectively
- Design, develop the PCB through Altium Designer and get it manufactured, providing all necessary files such as BOM, gerber files, etc.
- Solder all through hole and SMT components including masking for PCB and Prototype; create any needed jumper wires for connections through crimping and soldering
- Prototype implementation of a perf board circuit, also used as an emergency backup board for PCB failure ensuring continuation of product development
- Implement Sensors and Encoders a filter for the line Camera or velocity measured input through embedded software development and test through debugging microcontroller software

University of Illinois Chicago (UIC) Chicago, United States

Aug 2021 - May 2022

Senior Design Product Development

- Work in a four-student team to prototype a product implementing a automated watering systems plants by taking moisture levels, outdoor weather conditions, and plant information data into account
- Manage team through verbal and written communication to make sure all assignments and goals are completed on time, and submit weekly reports based on progress of project development
- Embedded programming for a Arduino Nano iot 33 to consider watering decisions per plants moisture levels and water consumption
- Create mobile APP (Kivy framework) to display necessary information regarding the system, plants moisture levels and weather conditions
- Create a UDP client-server connection for communication between powered product and the mobile application; providing manual control over a wireless connection
- Implement wireless communication through Bluetooth Protocol (BLE) for consumer and their device

Languages:

- Spanish (B-1) English (Advanced)